

VariantFit: Social Fitness Platform

Project Architecture & Cost Analysis Report

1. Project Overview

VariantFit is a modern, full-stack social fitness platform designed to help users track their workouts, manage diets, share their fitness journey through posts and immersive short-form videos (Reels), and connect with a community. It features real-time messaging, AI-driven diet planning, robust workout tracking, and a dynamic Instagram-like user interface.

2. Technology Stack

The application is built using a modern JavaScript/TypeScript ecosystem, ensuring high performance, scalability, and seamless user experience.

- **Frontend Framework:** Next.js 16 (React framework) utilizing App Router and Turbopack for rapid compilation.
- **Language:** TypeScript & JavaScript
- **Styling:** Tailwind CSS (Utility-first CSS framework for responsive and modern UI)
- **Icons:** Lucide React
- **Backend & Real-time Database:** Google Firebase (Firestore)
- **Authentication:** Firebase Authentication (Email/Password & Google OAuth)
- **Media Storage & CDN:** Cloudinary (Optimized image and video delivery) & Firebase Storage

3. Cloud Infrastructure & Storage

- **Database (Firestore):** A NoSQL document database used to store users, posts, comments, friend requests, notifications, and real-time chat messages.
- **Authentication State:** Managed securely via Firebase Auth, supporting robust session management and Google Sign-in.

- **Media Storage (Cloudinary):** Handles high-bandwidth assets like profile pictures, progress photos, and short-form videos (Reels). Cloudinary automatically optimizes videos for web playback and handles aspect-ratio cropping.
- **Deployment/Hosting:** Vercel (Next.js native hosting platform), providing edge-network caching and serverless functions for API endpoints.

4. Cost Analysis & Expected Expenses

For a college project, portfolio piece, or early-stage startup, VariantFit is designed to operate entirely on **Free Tiers**. Below is the breakdown of expected costs for development and early production:

Service	Purpose	Free Tier Limits	Estimated Cost
Vercel (Hobby Plan)	Web Hosting & API Edge Functions	100GB Bandwidth/month, unlimited deployments	\$0.00 / month
Firebase (Spark Plan)	Database, Auth, Cloud Storage	50K Document reads/day, 20K writes/day, 1GB total DB storage, 50K Monthly Active Users	\$0.00 / month
Cloudinary (Free Plan)	Video & Image Storage/Streaming	25 Monthly Credits (Approx. 25GB storage or 25GB bandwidth/month)	\$0.00 / month
Total Expenses	Overall Infrastructure	Sufficient for hundreds of active daily users	\$0.00 / month

** Note: Expenses will only occur if the application scales drastically to thousands of daily active users streaming videos simultaneously. For a college project, you will not exceed these free limits.*

5. Key Features Implemented

- **Immersive Social Feed & Reels:** TikTok/Instagram-style vertical video scrolling with auto-play, intersection observers, and double-tap to like.
- **Real-time Chat Engine:** Direct messaging with live updates, friend requests, and clickable shared post links.

- **Comprehensive Fitness Tracking:** Logging workout sets, reps, daily calories, and rotating AI-generated diet plans.
- **Privacy & Security:** Public/Private account toggling, friend-only visibility, and role-based access control (Admin vs User).